

Sixteen passes: what was proposed, what was tested, and what survived

Ava Kim — cat-pose-benchmark, technical note 06, v0.2, 20 September 2026 (v0.2: criteria in Table 1 corrected against the pass log after the reviewer's audit; passes 17–20 added)

Repository: <https://github.com/in-c0/cat-pose-benchmark> · Follows note 05 · Licence of this note: CC-BY-4.0

Abstract

This is the audit trail behind note 05. Over one night, twenty passes with a reviewer model produced sixteen proposals for improving the grounded tail method were made by a reviewer model, built, tested against a written-down criterion, and kept or dropped on the result. One survived: the two-sided SAM2 bridge (v3). Everything else — denser sampling, re-weighted temporal terms, a wider candidate lattice, a lower detector threshold, a looser gate, a cheaper switch, tracking until failure, a consensus audit, two forms of zero-shot semantic verification and one of mask geometry — failed the test it was given, and the table below says which test and by how much. The frame truth was revised four times on the way, and those revisions are in the same table, because three of them were found by method output and the record should say so. The pass log with every proposal and reply in full is in the portfolio hub. §5 is the reviewer's own closing reading: measure the oracle ceiling of the current stack on human curve truth before choosing the next component to change.

1. Why a table

Note 05 reports the outcome. It does not show how many things were tried and dropped to get there, and a reader deciding whether to try one of them again needs the negative results as much as the positive one. Each row below is one pass: what the reviewer proposed, what was built, the criterion written down before the result was known, the number, and the disposition. Where the criterion was not written down first, the row says so.

2. The passes

Table 1. The passes. Counts are TP / wrong on visible / missed / wrong on hidden on the four development clips under the truth version and scorer in force at the time, unless stated; note 05 v0.3 re-reports the surviving methods under scorer v2. "Rejected" means the method was not changed. The criteria in rows 2, 3, 5, 6a and 13 were corrected in v0.2 against the pass log after the reviewer pointed out that v0.1 had paraphrased them.

pass	hypothesis	what was built	predeclared test	result	disposition
1	denser sampling helps the temporal pass	inference at 2–4× the review rate, scored at the review timestamps	+recall, ≤ 1 new FP	+3 recall points under the old proxy; 73/1/25/8 vs 77/0/22/2 once the truth was written down	rejected; the proxy was the error
2	temporal terms should scale with frame interval	time-normalised Viterbi <i>emissions</i> (the switch cost stayed at 0.20), reviewer's weights	recall up with no new FP on either clip	<i>Cat Plays</i> collapses: 63/0/32/1	rejected
3	a wider candidate lattice recovers misses	oracle: correct box in top-5 but not top-2	≥ 4 new usable tails in K5■K2, ≥ 3 of them surviving the crop check	1 frame	not built
4	—	truth v1 frozen; scorer; rescoring of every method	—	v1 77/0/22/2 (P 0.975, baseline R 0.778)	
5	the detector threshold hides true tails	dump of every "tail" box down to 0.05 on missed and on hidden-tail frames	≥ 4 usable boxes below 0.20 on missed frames, ≥ 3 of them at ≥ 0.10	2 usable; 2.9 junk boxes ≥ 0.20 per hidden frame	rejected
6a	the cat gate is too tight	30 % containment instead of the centre rule	none predeclared — a diagnostic I had already started	identical to v1	rejected
6b	a cheaper switch lets tails in	switch cost 0.20 → 0.15/0.10/0.05/0.00	a plateau before the first FP	first FP arrives with the first TP	rejected

pass	hypothesis	what was built	predeclared test	result	disposition
6c	two-sided propagation can fill a gap safely	left / right / union / intersection bridges, gaps ≤ 3	≥ 2 TP, 0 FP over v2	intersection +4 TP, 0 FP; the others admit a 3× blob	kept as v3
7	SAM2 can track a tail through a dark prefix	long one-sided propagation oracle with two negative controls	$\geq 3/4$ dark frames tracked, no drift in the controls	4/4; controls fail safe (mask empties)	evidence for pass 8
8	track until evidence of failure	v4: whole-run tracking, one-sided from the edges, stop on empty / no centreline / area $\geq 2.5\times$	$\Delta TP \geq 3$, ΔFP hidden = 0, no wrong-part on review	+5 TP, +8 FP: SAM2 slides onto a black leg and never empties	rejected
9	consensus can audit accepted frames	v3a: replace each accepted mask by the intersection of its neighbours' propagations	$\Delta FP \leq -1$, $\Delta TP \geq 0$, no new wrong-part	-2 FP, +2 TP — both were whole-cat boxes the truth had accepted (revision 1.2)	candidate, not default (it shaped the truth it gained on)
10	low-confidence partials are a truth problem	primary stratum = visible/partial with identity confidence high/medium; blinded re-review with ± 1 s of video	licking strip resolves or is excluded	unresolved; excluded (revision 1.3)	truth rule
11	v3 generalises	prospective holdout #1: four clips, metadata-only draw, truth before inference	$\Delta R > 0$, $\Delta P \geq -0.02$, no new repeated failure	ΔR 0, ΔP -0.012, a rump read as a tail for 17 frames	not met
12	—	note 05 v0.1	—	—	published
13	an existing SigLIP class already beats "tail" on the rump	offline margin audit of the six crop scores on every accepted frame	engineering screen: remove most of the Taiwan run, no new repeated failure, ≤ 2 recall points lost	parameter-free tail $> \max(\text{others})$: -31 dev TP, -40 holdout TP (a hanging cream tail is "leg")	rejected
14	the crop, not the model, is the problem	re-score on SAM2-mask-tight crops	Larry $\geq 30/40$ improve; Taiwan $\geq 12/17$ keep a negative margin; ≤ 5 dev flips	1/40; 14/17; 9 flips; AUROC 0.38 $\rightarrow$ 0.18	rejected; zero-shot verification dropped
15	v3 generalises (second draw)	holdout #2: next eight clips in the frozen order, truth before inference	as pass 11	ΔR +0.034, ΔP -0.008, no new repeated failure	met ; precision 0.73
16	mask geometry separates the false assertions	six features on 191 TP / 46 FP across all three sets	direction replicates on all sets, pooled AUROC ≥ 0.80 , ≥ 2 FP modes	max AUROC 0.67; direction reverses between sets	rejected; single-threshold geometry dropped
17	—	note 06 v0.1	—	—	published
18	the frozen embedding contains tail-vs-not-tail signal the text heads miss	linear probe on SigLIP embeddings of the 237 accepted crops, clip- and run-weighted, leave-two-clips-out over the 11 clips that have accepted frames (55 folds), threshold at 98 % of weighted training TPs	held-out FP -25 % with TP loss ≤ 2 %, on ≥ 3 clips, no clip of ≥ 10 TPs losing > 1	TP 191 $\rightarrow$ 44, FP 46 $\rightarrow$ 15; six clips lose more than one	rejected (exploratory: labels not human-checked)
19	the record has holes a reviewer would find	reviewer audit of scorer, truth rules, notes and manifests	—	scorer judged the detector's box, not the curve; visible-without-reference scored any assertion correct; prose counts disagreed with the manifests	scorer v2; one generated aggregate; notes 05 v0.3 and 06 v0.2

pass	hypothesis	what was built	predeclared test	result	disposition
20	—	the reviewer's closing reading for the next investigator	—	\$5	recorded

Table 2. Truth revisions.

version	pass	change	found by
1	4	first written-down truth: visibility, identity confidence, condition, reference box; reviewer is a model	—
1.1	8	second cat's tail recorded (a lt_*); scorer instance-agnostic; the tuxedo tail on the scratcher, f007–f017 of the jumping clip, annotated for the first time	the oracle in pass 7 tracking the other cat
1.2	9	walking f006/f007 references replaced (they were whole-cat detector boxes); containment rule made one-way	the audit in pass 9
1.3	10	primary stratum requires identity confidence high/medium; identity column; licking f027–f035 re-reviewed with temporal context and left unresolved	the reviewer model's rule
scorer v2	19	the reported curve is judged, not the detector's box; a collapsed centreline is unusable_curve; a visible tail without a reference box needs a hand judgement; holdouts scored as frozen and adjudicated	the reviewer's audit
—	15	<i>Andra and Billy</i> f012–f014 moved to uncertain after v1 drew a mask where the blind reading said hidden	method output; resolved down, never up

3. What the table says

Three things. First, the only change that survived is the one that adds no threshold and no new model: it asks two propagations to agree and refuses when they do not. Second, the truth moved four times, three of them because a method found something the annotator had not, and each time the fix was to make the truth stricter or to exclude, never to accept the method's reading. Third, the failures cluster: every semantic and structural verifier tried was a per-frame test of a crop or a mask, and every one failed because true tails and false assertions overlap on whatever single axis was chosen — a hanging cream tail is what the zero-shot model calls a leg, a curled grey tail is as body-attached as a whole-cat mask. The reviewer model's own reading at the end of pass 16 was to stop tuning here, and that is what this note does. Its audit at pass 19 then found that the scorer had been judging the detector's box rather than the curve the method reports, which hid six collapsed centrelines; the corrected numbers are in note 05 v0.3 and do not change any disposition in Table 1. One limitation stands over every number: the 603 frames come from 16 clips and are not 603 independent observations — one clip supplies 40 near-identical easy tails, another a 17-frame repeated mistake — so the aggregate also reports per-clip counts, a macro average over clips and the number of distinct false runs (seven on holdout #2, not 27).

4. What comes next

The reviewer model's proposal for the next phase was a linear probe on the frozen SigLIP image embedding of the existing crop, trained on the 237 accepted predictions (191 true, 46 false) with each clip and each contiguous same-label run given equal weight, evaluated by exhaustive leave-two-clips-out, threshold chosen on training clips only as the highest that keeps 98 % of weighted training true positives, and promoted only if held-out false assertions fall by 25 % with true ones down no more than 2 %, on at least three different held-out clips, with no clip of ten or more true frames losing more than one. It was proposed for 16 clips and 120 folds;

only 11 clips have an accepted frame, so the executed experiment was 55 folds. It was run as an exploratory probe, because the labels are model-annotated and a human has not checked them (that item is filed and open), and it failed every clause: held-out true frames fell from 191 to 44 while false ones fell from 46 to 15. With one cat per clip, a probe trained on ten cats' tails does not recognise the eleventh's, which is note 04's result again at a larger scale. The frozen embedding is not sufficient at this data size; the sequence stops there rather than escalating to fine-tuning.

5. What the next investigator should do first

This section is the reviewer model's closing reading of the whole sequence, asked for as the twentieth and last pass and reproduced here as written (ChatGPT, 20 September 2026), with two light edits for house style. It is the reviewer's view, attributed as such, and I agree with it.

> The next investigator should not begin with another prompt, threshold, verifier or temporal rule. The first job is measurement. > > The present benchmark has established something narrower but useful. Grounding DINO can often point at an extended tail. SAM2 can often turn a good prompt into a usable tail mask and can bridge short gaps when independent propagation from both sides agrees. But the system does not know reliably whether a plausible elongated region is a tail. Across the two prospective holdouts, the dominant errors were not random detector noise: they were stable, visually plausible alternatives — rump, whole cat, forepaw or leaf, face edge, and once a person's face. Temporal consistency reinforces those mistakes because they are consistent. Zero-shot SigLIP, tighter crops, simple mask geometry and a linear probe on frozen SigLIP embeddings all failed predeclared tests. The evidence so far does not support another rule layered on top. > > Before training anything, a human should verify the existing truth. More importantly, the next data collection should finally annotate the object this project claims to benchmark: the continuous tail curve itself. A fresh set of cats should have per-instance visible/partial/hidden labels and human tail centrelines or masks, with ambiguous cases independently reviewed. The existing box truth is useful for asking whether an output is somewhere on the tail, but it cannot tell whether the reported curve has the right shape. > > **The one measurement that would change the diagnosis.** The most informative next experiment is the oracle ceiling of the current pipeline. On fresh human curve-level truth, retain every tail proposal produced by the current Grounding DINO stage before the v1 selection machinery removes candidates. Run the existing SAM2 segmentation and centreline extraction for every proposal. Then, using the ground truth only for analysis, allow an oracle to choose the best resulting curve on each frame, or NONE when no tail should be reported. This asks: if candidate selection were perfect, how good could the current detector + SAM2 + centreline stack possibly be? Measure the distance between each oracle-selected curve and the human curve, preferably with a scale-normalised curve metric rather than another box threshold, and retain the reason when the oracle cannot recover the tail: no usable detector proposal, segmentation failure, or centreline failure. Compare that oracle result with v3 on the same frames. That gap determines what should happen next. > > If the oracle is very good and v3 is much worse, the components already contain the answer and selection is the bottleneck; collect substantially more cats and train a tail-specific selector or verifier with clip- and animal-disjoint evaluation. The failed eleven-clip linear probe says that the present data are not enough; it does not show that supervised selection is impossible. If the oracle itself misses many visible tails because no proposal reaches the real tail, stop working on verification: the detector is the ceiling; compare or train a tail-specific detector on curve-annotated data. If a good proposal exists but SAM2 frequently expands to body, leg or background, segmentation is the ceiling; train or adapt a tail segmenter instead of trying to classify its failures afterwards. If the mask is correct but the extracted centreline is wrong or collapses, the representation is the ceiling; evaluate the tail directly as a mask or ordered spline and replace the current skeletonisation step. This oracle experiment should be performed before a new detector bake-off or any fine-tuning, because otherwise several components will change at once and a gain will not identify what was limiting the system. > > **What not to repeat.** The negative results in this note should be treated as evidence, not unfinished tuning. Do not repeat global detector-threshold lowering without a new reason: hidden-tail frames already produce abundant false tail proposals. Do not lower the Viterbi switch cost to rescue isolated tails: the first recovered frame arrived with the first new false positive. Do not assume a higher frame rate improves the temporal model: fixed-timestamp evaluation reversed that conclusion. Do not extend one-sided SAM2 tracking merely because its mask remains non-empty: it can track a hind leg confidently for several frames. Do not add another zero-shot negative prompt because one failure happens to resemble that category: true tails already score as leg, belly or nearly zero depending on the animal and scene. Do not expect a single area, attachment or elongation threshold to separate tails from false masks: the observed true and false modes occupy both ends of those axes. And do not train another verifier on the present model-annotated crops and call cross-frame

volume a sample size; the eleven active clips behaved like eleven domains, not 237 independent examples. > > The other lesson is methodological. Preserve prospective data. Freeze truth before inference. Keep post-inference adjudication separate from as-frozen scoring. Group errors by clip and contiguous run as well as by frame. When a method exposes a truth error, make the truth stricter or uncertain rather than silently moving it toward the prediction. And keep negative experiments; several of the most useful results in these twenty passes were proposals that failed. > > **Where the ceiling is today.** I would not yet call the current performance a model ceiling. It is a zero-shot-composition ceiling on the evidence collected so far. The surviving v3 bridge demonstrates that temporal information can recover short missing observations without another learned model or fitted threshold. The two prospective holdouts demonstrate that its larger limitation is semantic identity: deciding that a particular visible structure is a tail at all. The failed zero-shot and frozen-embedding verifiers show that this decision is not captured robustly by the representations and small sample available here. The next phase should therefore be data-first, not rule-first. Human curve truth plus the current-stack oracle ceiling would tell us whether the next dollar and hour belong to candidate selection, detection, segmentation or curve extraction. Until that measurement exists, choosing among those directions would be another guess.

That is where the twenty-pass sequence stops.

Acknowledgements

As for note 05: the passes were a back-and-forth between Claude Code and ChatGPT; the truth was annotated by the former; the human pass is pending.